Chapter 20 · Semester 3

OSCILLATIONS

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15 worked examples 4 simulations Exercises with answers

Oscillations: introduction

Oscillation (periodic motion)

is **the back-and-forth motion of an object about a fixed point**.

- a mass on the end of a spring (horizontal or vertical)
- atoms vibrating in a crystal
- a simple pendulum
- a child on a swing

The simplest oscillation is a mass on a spring sliding on a smooth horizontal surface (Figure 20.1). We assume the spring is light (its mass is ignored) and there is no friction.

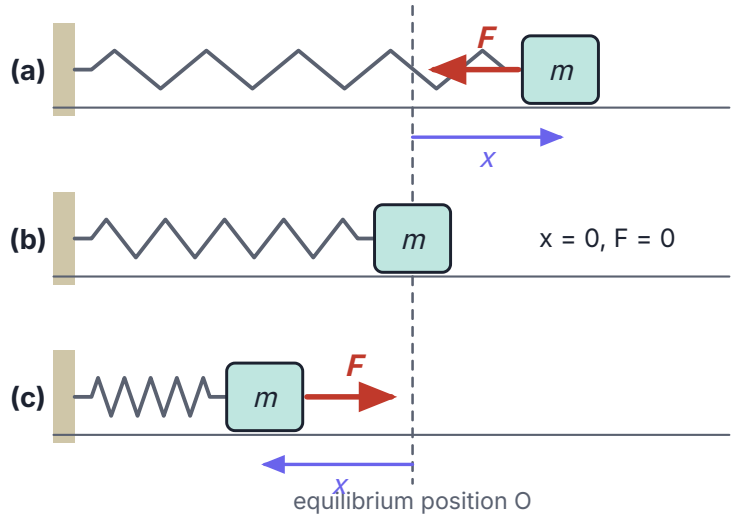


Figure 20.1: (a) spring stretched, the force acts to the left; (b) at the equilibrium position, F = 0; (c) spring compressed, the force acts to the right.

- The spring has a natural length at which it exerts no force on the mass. The position of the mass at this point is the **equilibrium position**.
- If the mass is pulled to the right (spring stretched) or pushed to the left (spring compressed), the spring exerts a force that acts to return it to equilibrium. This is the **restoring force**.
- At $x = 0$ the restoring force is zero, but the mass keeps moving past O because of its **inertia**.
- The restoring force is always directed towards the equilibrium position.

20.1 Characteristics of simple harmonic motion

Equation of SHM

By Hooke's law, the restoring force varies with the displacement x as

$$F = -kx$$

k = spring constant (N m^{-1}), a measure of the stiffness of the spring. The minus sign shows that F is always opposite to the displacement x.

With no friction this is the net force, so by Newton's second law, $F = ma$:

$$ma = -kx$$

$$a = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x$$

Since k/m is a positive constant, write it as ω^2:

$$a = -\omega^2 x$$

ω = angular frequency (rad s^{-1}). Acceleration $\propto$ displacement, and is opposite in direction to the displacement.

Simple harmonic motion (SHM)

is the periodic motion of an object in which **the acceleration (and the restoring force) is directly proportional to the displacement from a fixed point (the equilibrium position), and is always directed towards that point**.

Teacher's note

Whenever a question gives an equation of motion such as $\mathrm{d}^2x/\mathrm{d}t^2 + 100x = 0$, rewrite it as $a = -\omega^2 x$ first. Here $a = -100x$, so $\omega^2 = 100$ and $\omega = 10$ rad s^{-1}. Every other SHM quantity follows from ω.

20.2 Kinematics of simple harmonic motion

The reference circle

> **Reference circle**
> is the circle in which a particle moves at constant angular velocity so that **its projection onto a diameter of the circle performs SHM**.

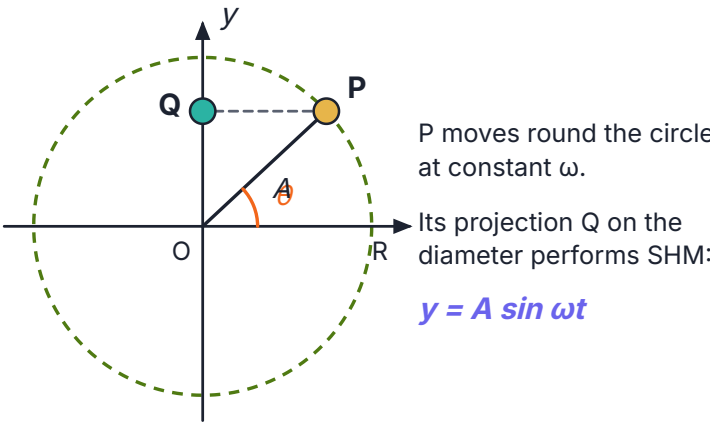


Figure 20.2: P moves anticlockwise round a circle of radius A at constant ω. Its projection Q moves up and down the diameter in SHM.

If OQ = y and θ = ωt, then $y = A \sin \theta$, so

$$y = A \sin \omega t \quad \text{or} \quad x = A \sin \omega t$$

This is why ω is called the **angular frequency**, even though the oscillating mass moves only in a straight line.

Terms used in SHM

Quantity	Meaning
Displacement, x (m)	the distance of the particle from the equilibrium position (with direction)
Amplitude, A (m)	the maximum magnitude of the displacement
Period, T (s)	the time for one complete oscillation
Frequency, f (Hz)	the number of complete oscillations per second; $f = \frac{1}{T}$
Angular frequency, ω (rad s^{-1})	$\omega = 2\pi f = \frac{2\pi}{T}$

Solutions of $a = -\omega^2 x$

Both of these satisfy $a = -\omega^2 x$. The only difference is where the clock starts ($t = 0$).

$x = A \sin \omega t$	$x = A \cos \omega t$
starts at the equilibrium position ($x = 0$), moving at maximum speed	starts at the maximum displacement ($x = A$), momentarily at rest
check: $v = \mathrm{d}x/\mathrm{d}t = A\omega \cos \omega t$; $a = \mathrm{d}v/\mathrm{d}t = -A\omega^2 \sin \omega t = -\omega^2 x$ ✓	check: $v = -A\omega \sin \omega t$; $a = -A\omega^2 \cos \omega t = -\omega^2 x$ ✓

A mass pulled to $x = +A$ and released from rest follows $x = A \cos \omega t$. A mass given a push through the equilibrium position at $t = 0$ follows $x = A \sin \omega t$.

Velocity and acceleration

Velocity (for $x = A \sin \omega t$)

$$v = \frac{\mathrm{d}x}{\mathrm{d}t} = A\omega \cos \omega t$$

Maximum when $\cos \omega t = \pm 1$:

$$v_{\max} = A\omega$$

Acceleration

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = -A\omega^2 \sin \omega t = -\omega^2 x$$

Maximum magnitude when $\sin \omega t = \pm 1$:

$$a_{\max} = A\omega^2$$

Velocity in terms of displacement

$$x = A \sin \omega t \Rightarrow \sin \omega t = \frac{x}{A}$$

$$\sin^2 \omega t + \cos^2 \omega t = 1 \Rightarrow \cos \omega t = \pm\sqrt{1 - \frac{x^2}{A^2}}$$

$$v = A\omega \cos \omega t = \pm A\omega\sqrt{1 - \frac{x^2}{A^2}} = \pm\omega\sqrt{A^2 - x^2}$$

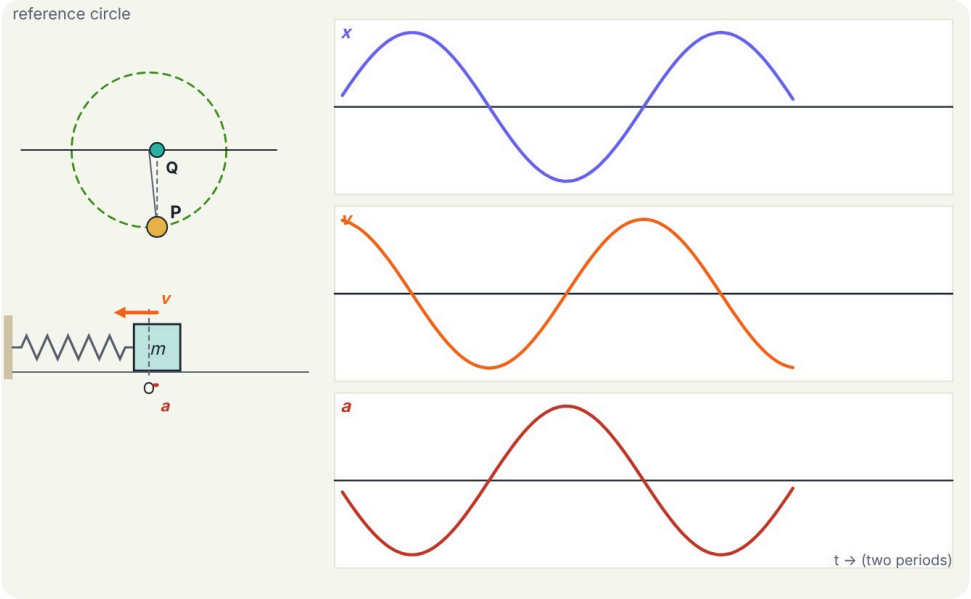
$$v = \pm\omega\sqrt{A^2 - x^2}$$

$\pm$: the particle can pass through the same point x in either direction. At $x = 0$, $v = \pm\omega A$ (maximum speed); at $x = \pm A$, $v = 0$.

Simulation

SHM and the reference circle

P goes round the circle at constant ω; its shadow Q (and the mass on the spring) moves in SHM. Watch the x–t, v–t and a–t graphs being drawn together. Change the amplitude and the period, and choose whether the clock starts at the equilibrium position or at the maximum displacement.



t	ω = 2π/T	x	v	a = −ω²x
2.97 s	3.14 rad/s	1.1 cm	−0.312 m/s	−0.104 m/s²

K (m = 0.5 kg)	U	E = K + U
24.4 mJ	0.3 mJ	24.7 mJ

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.2.1

A particle vibrates according to the equation $y = 4 \sin 3t$, where y is in metres and t in seconds. Calculate (a) the amplitude, (b) the angular frequency, (c) the frequency, (d) the period, (e) the maximum speed, (f) the maximum magnitude of the acceleration.

Solution

Compare with $y = A \sin \omega t$.

(a) A = **4 m** (b) ω = **3 rad s^{-1}**

(c) $f = \dfrac{\omega}{2\pi} = \dfrac{3}{2\pi}$ = **0.48 Hz** (d) $T = \dfrac{1}{f}$ = **2.09 s**

(e) $v_{max} = A\omega = (4)(3)$ = **12 m s^{-1}**

(f) $a_{max} = A\omega^2 = (4)(3)^2$ = **36 m s^{-2}**

Example 20.2.2

A 300 g mass at the end of a spring oscillates with an amplitude of 7.0 cm and a frequency of 1.80 Hz. When it is 3.0 cm from its equilibrium position, calculate (a) its speed, (b) the magnitude of its acceleration.

Solution

$\omega = 2\pi f = 2\pi(1.80) = 11.3$ rad s^{-1}

(a) $v = \omega\sqrt{A^2 - x^2} = 11.3\sqrt{0.070^2 - 0.030^2}$ = **0.71 m s^{-1}**

(b) $a = \omega^2 x = (11.3)^2(0.030)$ = **3.8 m s^{-2}**

Example 20.2.3

The equation of motion of a mass at the end of a spring is $y = 0.30 \cos 0.50t$ (in metres). Find the displacement, velocity and acceleration of the mass at (a) $t = 0$, (b) $t = 3.0$ s.

Solution

$v = \frac{dy}{dt} = -(0.30)(0.50) \sin 0.50t = -0.15 \sin 0.50t$; $a = \frac{dv}{dt} = -(0.30)(0.50)^2 \cos 0.50t = -0.075 \cos 0.50t$

(a) $t = 0$: $y =$ **0.30 m**, $v =$ **0**, $a =$ **$-0.075\ \text{m s}^{-2}$**

(b) $t = 3.0$ s (calculator in radians): $y = 0.30 \cos 1.5 =$ **0.021 m**; $v = -0.15 \sin 1.5 =$ **$-0.15\ \text{m s}^{-1}$**; $a = -0.075 \cos 1.5 =$ **$-5.3 \times 10^{-3}\ \text{m s}^{-2}$**

STPM 2023 · Section A · Q2

A particle undergoes simple harmonic motion with a period of 2.0 s and amplitude 2.4 cm. What is the velocity of the particle when the displacement is 1.2 cm?

A. $6.5 \times 10^{-2}\ \text{m s}^{-1}$

B. $3.8 \times 10^{-2}\ \text{m s}^{-1}$

C. $2.1 \times 10^{-2}\ \text{m s}^{-1}$

D. $1.3 \times 10^{-2}\ \text{m s}^{-1}$

Answer

Answer: A

$\omega = 2\pi/T = \pi$ rad s^{-1}; $v = \omega\sqrt{A^2 - x^2} = \pi\sqrt{0.024^2 - 0.012^2} = 6.5 \times 10^{-2}$ m s^{-1}.

Common mistake: using T as ω, or forgetting to change cm to m.

Graphs of SHM against time

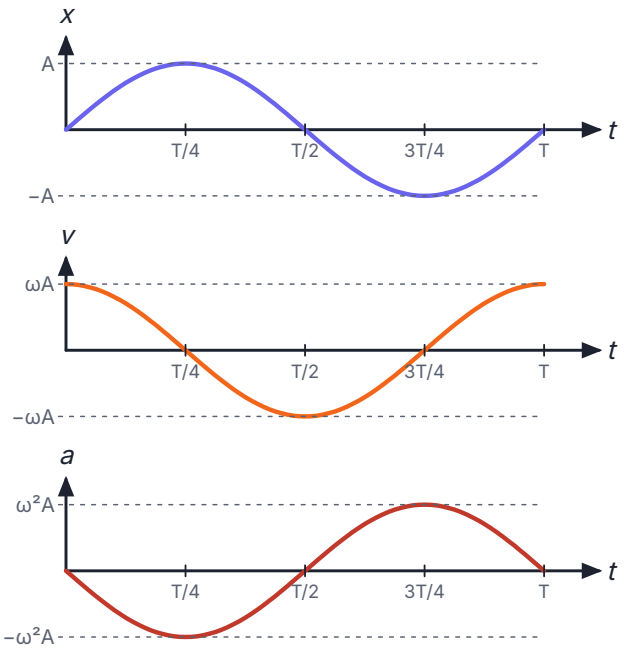
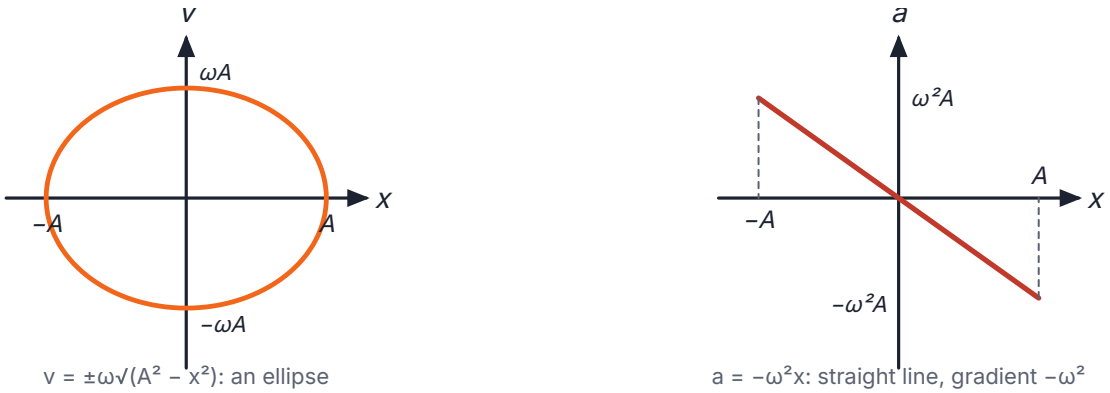


Figure 20.3: for x = A sin ωt, the velocity leads the displacement by a quarter period (T/4), and the acceleration is always opposite to the displacement.

t	x	v	a
0	0	$+\omega A$	0
$T/4$	$+A$	0	$-\omega^2 A$
$T/2$	0	$-\omega A$	0
$3T/4$	$-A$	0	$+\omega^2 A$
T	0	$+\omega A$	0

- When x is at a maximum or a minimum, $v = 0$ and $\lvert a \rvert$ is maximum.
- When $x = 0$ (passing through equilibrium), $\lvert v \rvert$ is maximum and $a = 0$.

Graphs of velocity and acceleration against displacement



v = ±ω√(A² − x²): an ellipse

a = −ω²x: straight line, gradient −ω²

Figure 20.4: v against x is an ellipse; a against x is a straight line through the origin with gradient −ω².

Since $F = ma = -m\omega^2 x$, the graph of restoring force against displacement is also a straight line through the origin. Its gradient has magnitude $k = m\omega^2$.

Example 20.2.4

The equation of motion of a particle in SHM is $x = 5 \sin 3t$, where x is in cm. (a) What is its amplitude? (b) Find its period. (c) Find its displacement at $t = 0.2$ s. (d) Sketch the graph of displacement against time.

Solution

(a) A = **5 cm**

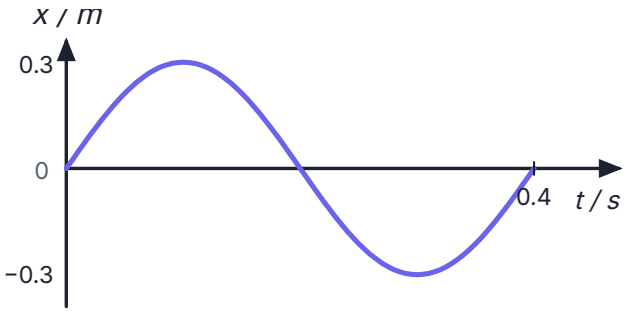
(b) $\omega = 3 = 2\pi/T \Rightarrow T$ = **2.09 s**

(c) $x = 5 \sin 3(0.2) = 5 \sin 0.6$ = **2.82 cm**

(d) a sine curve starting at 0, peak 5 cm at $t = 0.52$ s, back to 0 at 1.05 s, −5 cm at 1.57 s, 0 at 2.09 s.

Example 20.2.5

The displacement–time graph of an oscillating object is shown. Determine (a) the amplitude, the period and the frequency, (b) the angular frequency, (c) the equation of displacement as a function of time, (d) the equations of velocity and acceleration as functions of time.



Solution

(a) A = **0.3 m**, T = **0.4 s**, $f = 1/T$ = **2.5 Hz**

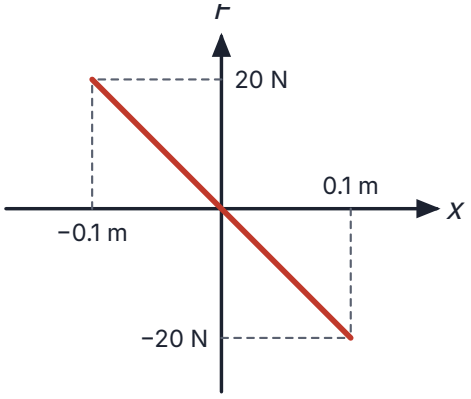
(b) $\omega = 2\pi f$ = **5π rad s^{-1}** (15.7 rad s^{-1})

(c) at t = 0, x = 0 and increasing, so x **= 0.3 sin 5πt**

(d) $v = \mathrm{d}x/\mathrm{d}t$ = **1.5π cos 5πt** m s^{-1}; $a = \mathrm{d}v/\mathrm{d}t$ = **−7.5π^2 sin 5πt** m s^{-2}

Example 20.2.6

The graph shows the force F acting on a particle of mass 2 kg against its displacement x. What type of motion is the particle following? Give a reason. What is its amplitude? Find (i) its angular frequency, (ii) its period, (iii) its maximum speed.



Solution

SHM, because the force is directly proportional to the displacement and opposite in direction to it (a straight line through the origin with negative gradient).

Amplitude = **0.1 m**

(i) $|\text{gradient}| = m\omega^2$: $\frac{20}{0.1} = 2\omega^2 \Rightarrow \omega =$ **10 rad s^{-1}**

(ii) $T = 2\pi/\omega =$ **0.63 s** (iii) $v_{max} = A\omega = (0.1)(10) =$ **1.0 m s^{-1}**

20.3 Energy in simple harmonic motion

Kinetic energy

$K = \frac{1}{2}mv^2$ and $v^2 = \omega^2(A^2 - x^2)$

$$K = \frac{1}{2}m\omega^2(A^2 - x^2)$$

maximum at $x = 0$: $K_{max} = \frac{1}{2}m\omega^2A^2$; zero at $x = \pm A$. In terms of time ($x = A \sin \omega t$): $K = \frac{1}{2}m\omega^2A^2 \cos^2 \omega t$

Potential energy

Work done against the restoring force: $U = \int_0^x kx\,\mathrm{d}x = \frac{1}{2}kx^2$

$k = m\omega^2$

$$U = \frac{1}{2}m\omega^2 x^2$$

zero at $x = 0$; maximum at $x = \pm A$: $U_{max} = \tfrac{1}{2}m\omega^2 A^2$. In terms of time: $U = \tfrac{1}{2}m\omega^2 A^2 \sin^2 \omega t$

Total energy is conserved

$E = K + U = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{1}{2}m\omega^2 x^2$

$$E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2 = \text{constant}$$

the total mechanical energy of SHM is conserved if no external force acts on it. $E \propto A^2$: doubling the amplitude makes the energy four times larger.

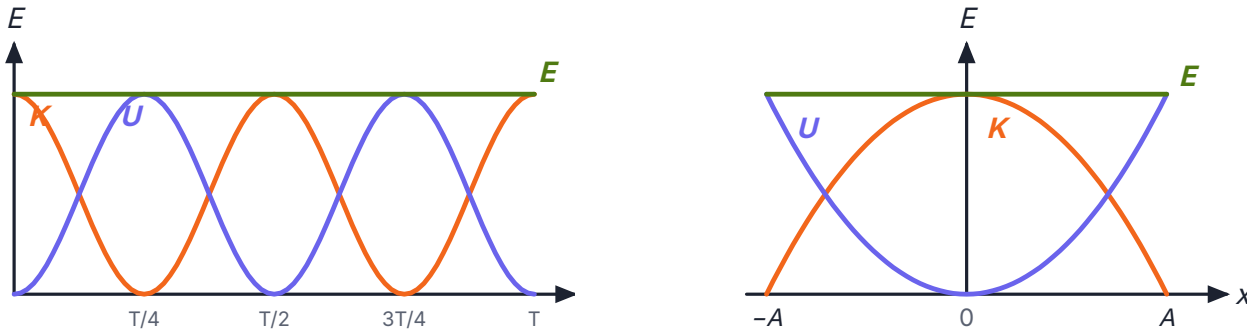


Figure 20.5: left, K and U against time (they vary at twice the frequency of x); right, K and U against displacement (two parabolas). E is constant in both.

t	x	K	U
0	A	0	$\frac{1}{2}kA^2$
$T/4$	0	$\frac{1}{2}kA^2$	0
$T/2$	$-A$	0	$\frac{1}{2}kA^2$
$3T/4$	0	$\frac{1}{2}kA^2$	0
T	A	0	$\frac{1}{2}kA^2$

(This table starts from $x = A$, i.e. $x = A\cos\omega t$.)

Misconception	Correct idea
Total energy changes as the object oscillates.	$E = \frac{1}{2}m\omega^2A^2$ is constant. Only K and U change.
Kinetic energy is greatest at the extremes.	K is greatest at $x = 0$ and zero at $x = \pm A$.
K–t and U–t graphs look like the x–t graph.	They vary at twice the frequency of x and are never negative.

Example 20.3.1

A 1.0 kg mass vibrates according to $y = 0.42\cos 7.4t$ (y in m, t in s). Determine (a) the amplitude, (b) the frequency, (c) the total energy, (d) the kinetic energy and the potential energy when $y = 0.16$ m.

Solution

(a) A = **0.42 m** (b) $f = \omega/2\pi = 7.4/2\pi$ = **1.2 Hz**

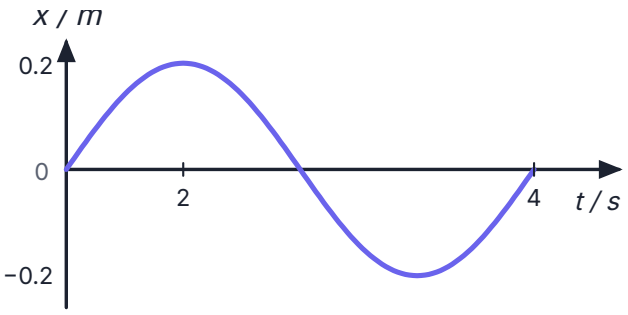
(c) $E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}(1.0)(7.4)^2(0.42)^2$ = **4.8 J**

(d) $v = \omega\sqrt{A^2 - y^2} = 7.4\sqrt{0.42^2 - 0.16^2}$ = 2.87 m s^{-1}; $K = \frac{1}{2}(1.0)(2.87)^2$ = **4.1 J**

$U = E - K = 4.8 - 4.1$ = **0.7 J**

Example 20.3.2

The displacement–time graph of an oscillating object of mass 0.3 kg is shown. Calculate the total energy (kinetic + potential) of the object.



Solution

A = 0.2 m, T = 4 s ⇒ ω = 2π/T = π/2 rad s^{-1}

$E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}(0.3)(\pi/2)^2(0.2)^2$ = **0.015 J**

STPM 2022 · Section A · Q1

An object of mass 0.10 kg undergoes SHM along the x-axis according to the equation $d^2x/dt^2 + 100x = 0$. If the total energy of the system is 0.0020 J, what is the amplitude of the motion?

A. 0.040 cm

B. 0.20 cm

C. 0.40 cm

D. 2.0 cm

Answer

Answer: D

$d^2x/dt^2 = -100x \Rightarrow \omega^2 = 100$.

$A^2 = 2E/(m\omega^2) = 2(0.0020)/(0.10 \times 100) = 4.0 \times 10^{-4}$ m^2 ⇒ A = 0.02 m = 2.0 cm.

STPM 2023 · Section A · Q1

Which statement is true about simple harmonic motion?

A. The acceleration is uniform.

B. The amplitude decreases with time.

C. The restoring force dissipates naturally.

D. The kinetic and potential energies interchange continuously.

Answer

> **Answer: D**
>
> A: the acceleration varies ($a = -\omega^2 x$). B and C describe **damped** oscillation, not ideal SHM. D: the total energy is constant while K and U trade places.
>
> **Tip:** "true about SHM" questions often hide a damped-oscillation distractor.

Exercises: kinematics and energy of SHM

1. A particle moves in SHM according to $y = 10 \sin \omega t$ m. If the period is 30 s, find (a) the amplitude, (b) the maximum velocity, (c) the maximum acceleration, (d) the displacement, velocity and acceleration when $t = 15$ s.

   Answer

   (a) 10 m (b) 2.1 m s^{-1} (c) 0.44 m s^{-2} (d) at $t = T/2$: $y = 0$, $v = -2.1$ m s^{-1}, $a = 0$

2. An object moving with SHM has an amplitude of 2 cm and a frequency of 20 Hz. Calculate (a) the period, (b) the acceleration at the middle and at the end of an oscillation, (c) the velocities at the same two points.

   Answer

   (a) 0.05 s (b) middle: 0; end: 316 m s^{-2} (c) middle: 2.5 m s^{-1}; end: 0

3. A block of mass 0.25 kg on a frictionless surface is attached to a light spring of spring constant 180 N m^{-1}. It is displaced 15 cm from equilibrium and released. Find (a) the total energy of the system, (b) the speed of the block when it is 10 cm from equilibrium.

   Answer

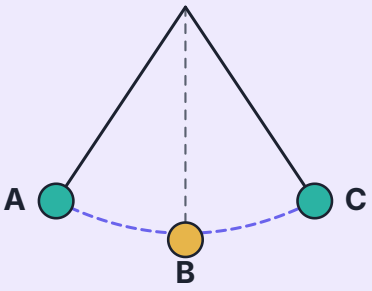
   (a) 2.0 J (b) 3.0 m s^{-1}

4. A 50 g mass vibrates in SHM at the end of a spring. The amplitude is 12 cm and the period is 1.70 s. Calculate (a) the frequency, (b) the spring constant, (c) the maximum speed, (d) the maximum acceleration, (e) the speed when the displacement is 6.0 cm, (f) the acceleration when $x = 6.0$ cm.

   Answer

   (a) 0.588 Hz (b) 0.68 N m^{-1} (c) 0.44 m s^{-1} (d) 1.6 m s^{-2} (e) 0.38 m s^{-1} (f) −0.82 m s^{-2}

5. The diagram shows a pendulum oscillating between A and C; B is the lowest point. State all the positions where (a) the acceleration is maximum, (b) the acceleration is zero, (c) the speed is maximum, (d) the velocity is zero, (e) the kinetic energy is maximum, (f) the potential energy is maximum.



Answer

(a) A and C (b) B (c) B (d) A and C (e) B (f) A and C

20.4 Systems in simple harmonic motion

Spring–mass system

Hooke's law and Newton's second law: $ma = -kx \Rightarrow a = -\frac{k}{m}x$

Compare with $a = -\omega^2 x$: $\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$

$T = \frac{2\pi}{\omega}$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

a larger mass gives a longer period; a stiffer spring (larger k) gives a shorter period. T does not depend on the amplitude.

For a **vertical** spring, gravity only shifts the equilibrium position down by mg/k. The mass oscillates about this new equilibrium with the same period, $T = 2\pi\sqrt{(m/k)}$. g does not appear in the formula.

Example 20.4.1

A mass of 200 g is attached to a horizontal spring on a frictionless surface. The mass is pulled 2.00 cm to the right of its equilibrium position and released. If the spring constant is 100 N m^{-1}, find the frequency of oscillation.

Solution

$\omega^2 = k/m$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{100}{0.200}} = \mathbf{3.56\ Hz}$$ (the 2.00 cm amplitude is not needed)

Example 20.4.2

A 0.50 kg mass is attached to a spring of spring constant 20 N m^{-1} and set oscillating. Find the period. What happens to the period if the spring is replaced by one four times as stiff?

Solution

$T = 2\pi\sqrt{0.50/20} = 2\pi(0.158) = \mathbf{0.99\ s}$

$T \propto 1/\sqrt{k}$, so with k = 80 N m^{-1} the period **halves** (0.50 s).

Simple pendulum

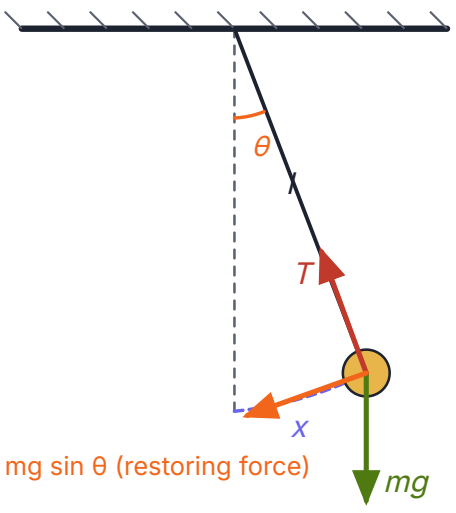


Figure 20.6: the component mg sin θ along the arc pulls the bob back towards the equilibrium position.

Restoring force along the arc: $F = -mg \sin\theta$

For small angles ($\theta < 10°$), $\sin\theta \approx \theta = \frac{x}{l}$ (θ in radians)

$$ma = -mg\frac{x}{l} \Rightarrow a = -\frac{g}{l}x$$

Compare with $a = -\omega^2 x$: $\omega = \sqrt{\frac{g}{l}}$

$$T = 2\pi\sqrt{\frac{l}{g}}$$

depends only on the length l and on g, not on the mass of the bob or (for small angles) on the amplitude

	Spring–mass system	Simple pendulum
Period	$T = 2\pi\sqrt{(m/k)}$	$T = 2\pi\sqrt{(l/g)}$
Depends on	mass m, spring constant k	length l, gravitational acceleration g
Does not depend on	amplitude A (and g)	mass of the bob, amplitude (small angles)

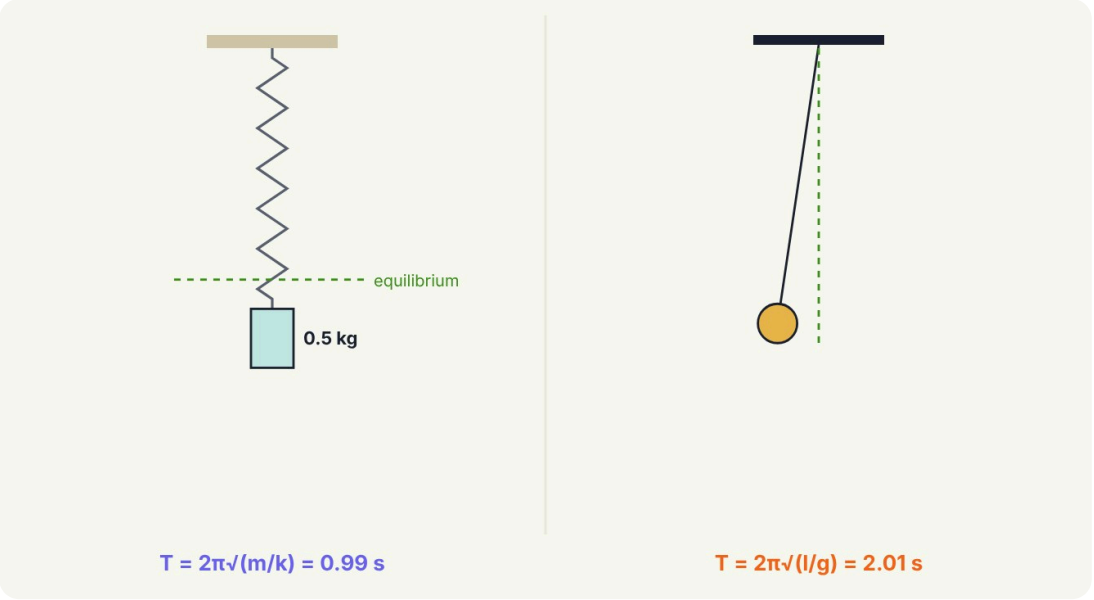
Teacher's note

A heavier phone on a spring bounces more slowly, but a heavier pendulum bob swings at exactly the same rate. Always check which system the question describes before you choose the formula. On the Moon (smaller g) a pendulum clock runs slow; a spring–mass system keeps the same period.

Simulation

Spring–mass system and simple pendulum

Change the mass, spring constant, length and g, and compare the two periods. Change the amplitude too: the period does not change. Try the Moon button.



spring–mass period	spring–mass frequency	pendulum period	pendulum frequency
0.99 s	**1.01 Hz**	**2.01 s**	**0.50 Hz**

the pendulum does not depend on **mass, amplitude**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.4.3

A simple pendulum has a length of 1.00 m. Taking $g = 9.81\ \text{m s}^{-2}$, find its period. What length gives a period of 1.0 s?

Solution

$T = 2\pi\sqrt{1.00/9.81} = \mathbf{2.01\ s}$

$l = g(T/2\pi)^2 = 9.81(1.0/2\pi)^2 = \mathbf{0.25\ m}$

Example 20.4.4

A student uses a simple pendulum of length 80.0 cm to determine g. There are 20 oscillations in 35.9 s. Find (a) the value of g, (b) the period if the experiment is done on the Moon, where the gravitational field strength is 1/6 of that on Earth.

Solution

T = 35.9/20 = 1.795 s

(a) $T^2 = 4\pi^2 \frac{l}{g} \Rightarrow g = \frac{4\pi^2(0.800)}{(1.795)^2}$ = **9.80 m s^{-2}**

(b) $T^2 g$ = constant: $T^2(g/6) = (1.795)^2 g \Rightarrow T = 1.795\sqrt{6}$ = **4.40 s**

20.5 Damped oscillation

Damping

is **the decrease in amplitude of an oscillation caused by dissipative forces** such as friction and air resistance.

- Part of the energy of the system is used to overcome these resistive forces, and is lost as heat.
- So the amplitude does not stay constant: it decreases gradually (exponentially) until the oscillation stops. Since $E \propto A^2$, the energy decreases too.
- A damped oscillation is therefore not perfect SHM.

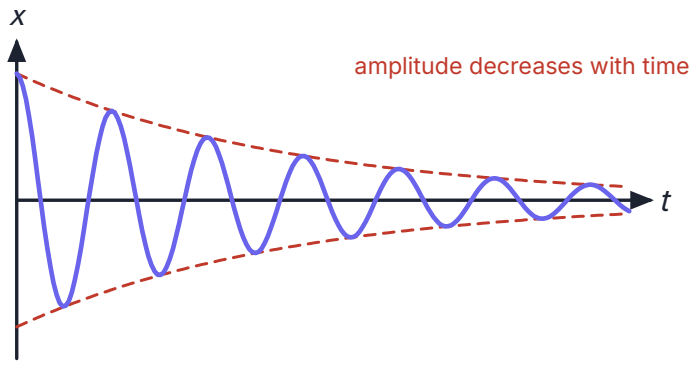


Figure 20.7: under damping: the oscillation continues but the amplitude decreases with time.

Three types of damping

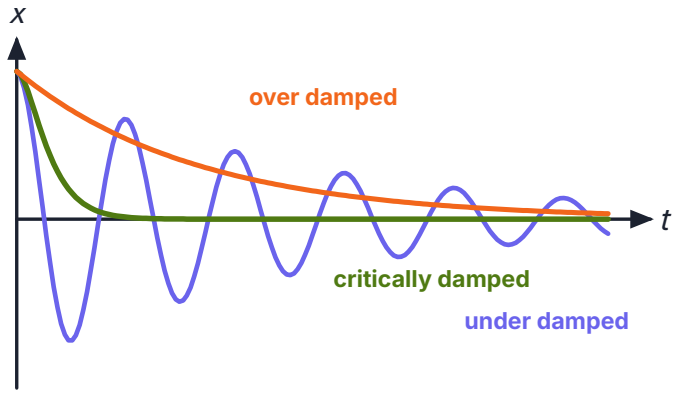


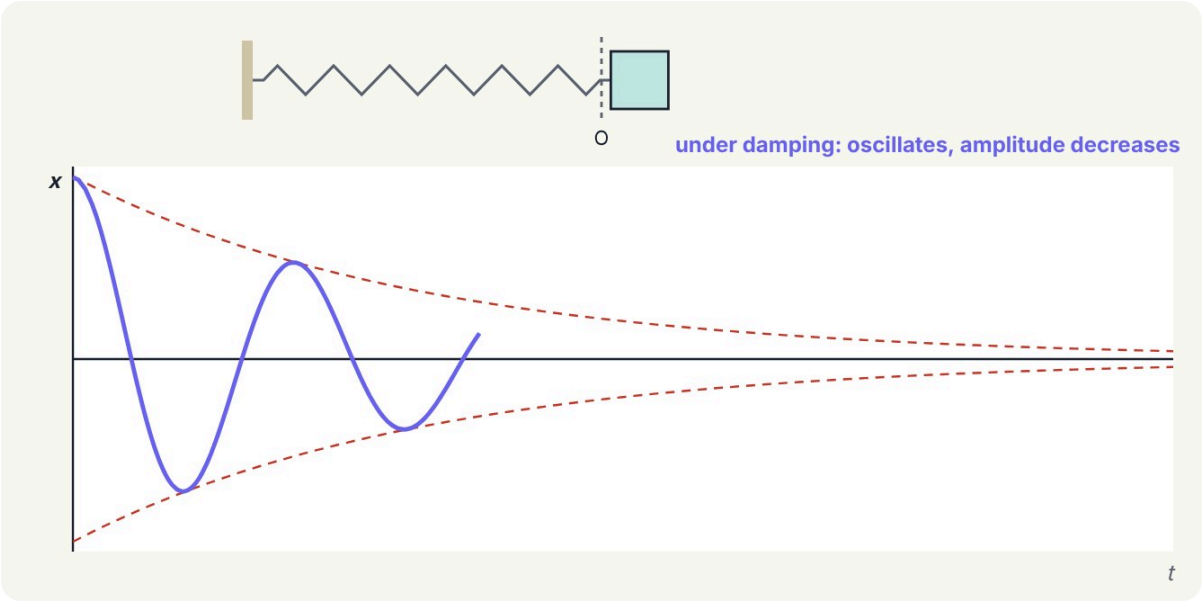
Figure 20.8: displacement against time for the three types of damping, each released from the same displacement.

Type	What happens	Example
Under damping (light damping)	The system oscillates, with the amplitude decreasing with time until it becomes zero. It takes a long time to die away.	a mass oscillating in air; a plucked guitar string
Critical damping	The system does not oscillate. It returns to the equilibrium position in the **shortest possible time**.	the shock absorbers of a car
Over damping (heavy damping)	The system does not oscillate, but it takes a **long time** to return to equilibrium.	a car fuel gauge; bathroom scales; a heavy door closer

Simulation

Damped oscillation

Increase the damping from none to under, critical and over damping. Watch the mass, the displacement–time graph and the energy. Critical damping brings the mass back to O fastest without overshooting.



time	displacement	energy left
3.0 s	0.14 × A_o	9 %

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

20.6 Forced oscillation and resonance

Free oscillation	Forced oscillation
The system is displaced and released, then left alone: no driving force.	A periodic external force (the driving force) acts on the system continuously.
It oscillates at its own **natural frequency** f_0.	After a short time, it oscillates at the **driving (forcing) frequency** f, not at f_0.
Its amplitude depends only on the energy given at the start (it is damped gradually if there is resistance).	Its amplitude depends on how close f is to f_0.

Natural frequency

is **the frequency at which a system oscillates when there is no driving force and no damping**. For a simple pendulum, $f_0 = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$.

Barton's pendulums

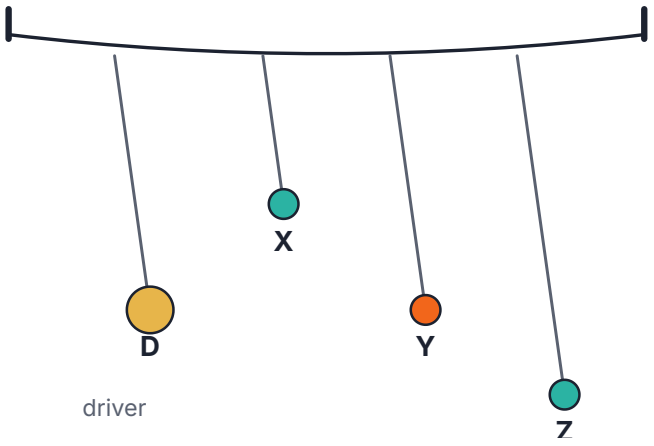


Figure 20.9: the driver pendulum D and pendulums X, Y, Z hang from the same string. Only Y has the same length as D.

- X, Y and Z are at rest at first. D is set swinging.
- Energy is passed along the string, so X, Y and Z start to oscillate (forced oscillation), all at the frequency of D.
- Y, with the same length (and so the same natural frequency) as D, oscillates with the **largest amplitude**. Y is in **resonance**.

Resonance

occurs when **the frequency of the driving force is equal to the natural frequency of the system**. The system then absorbs energy from the driving force most efficiently, and the amplitude of oscillation is maximum.

resonance: $f = f_0$

the condition for resonance. This frequency is the resonance frequency.

Resonance and damping

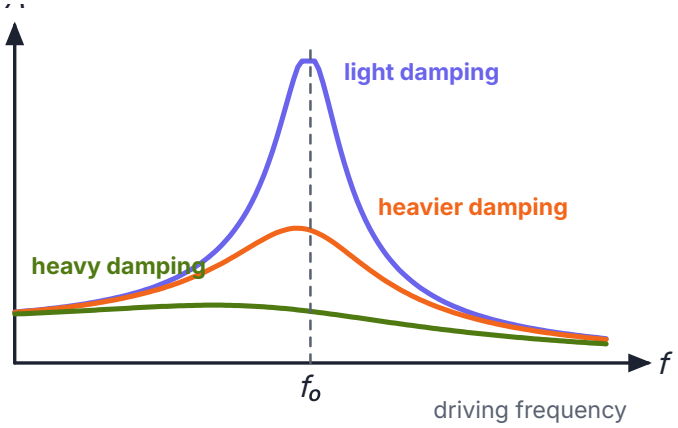
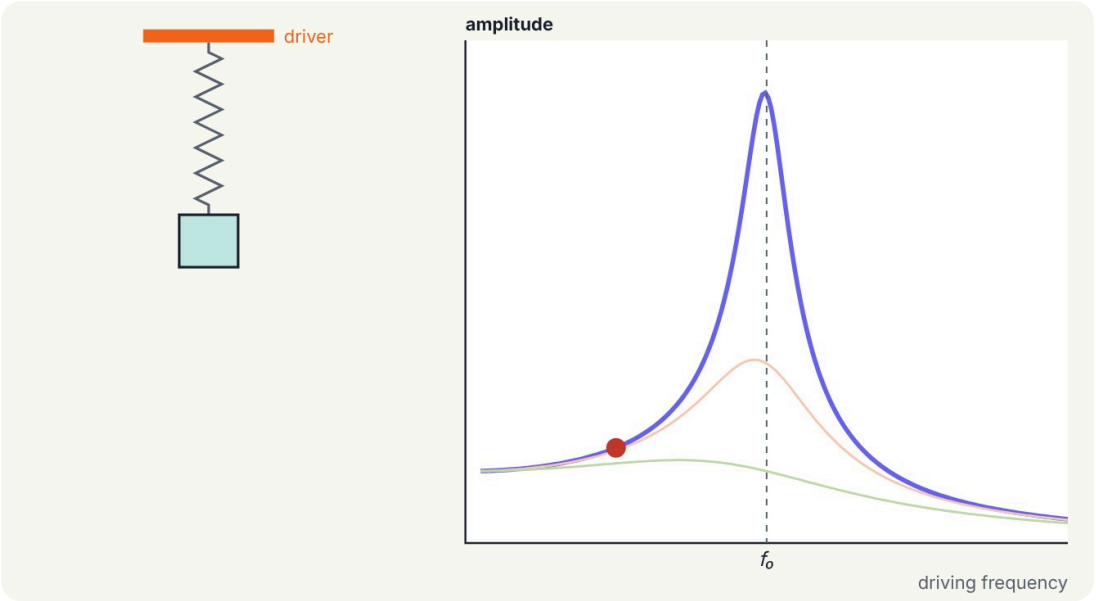


Figure 20.10: amplitude of the forced oscillation against driving frequency for different amounts of damping.

- Damping absorbs energy and reduces the effect of resonance.
- As the damping increases, the amplitude at resonance decreases and the resonance peak becomes broader (flatter).
- With heavier damping, the peak also moves slightly below f_0.

Simulation Forced oscillation and resonance

Slowly change the driving frequency and watch the amplitude of the driven mass. The amplitude is largest when the driving frequency equals the natural frequency. Then increase the damping and see the peak fall and become broader.



natural frequency f_o	driving frequency f	steady amplitude	the mass oscillates at
1.00 Hz	**0.50 Hz**	**1.3 × driver**	**0.50 Hz (= f)**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Examples of resonance

Destructive

- 1940, Washington State, USA: the Tacoma Narrows suspension bridge collapsed about four months after opening. A strong wind drove the bridge into large oscillations.
- 1850, Angers, France: more than 200 soldiers died when a bridge they were marching over collapsed. This is why soldiers break step when they march across a bridge.
- A singer can shatter a wine glass with a note at its natural frequency.

Useful

- The tuning circuit of a radio or TV resonates at the frequency of the chosen station.
- Wind instruments produce sound by making the air column inside resonate.
- Microwave ovens heat food because water molecules absorb energy from the microwaves.

Teacher's note

Questions often ask for the condition for resonance. Write it as a full sentence: "the driving frequency is equal to the natural frequency of the system".

Summary

Subtopic	Key results
20.1	$F = -kx$; $a = -\omega^2 x$
20.2	$x = A \sin \omega t$ or $A \cos \omega t$; $v = \pm\omega\sqrt{(A^2 - x^2)}$; $v_{max} = \omega A$; $a_{max} = \omega^2 A$
20.3	$K = \frac{1}{2}m\omega^2(A^2 - x^2)$; $U = \frac{1}{2}m\omega^2 x^2$; $E = \frac{1}{2}m\omega^2 A^2$ (constant)
20.4	$T = 2\pi\sqrt{(m/k)}$ (spring–mass); $T = 2\pi\sqrt{(l/g)}$ (simple pendulum)
20.5	amplitude and energy decrease; under, critical and over damping
20.6	resonance when the driving frequency = natural frequency

- Always match the given equation to $a = -\omega^2 x$ first to find ω.
- Keep every quantity in SI units (m, s) before substituting.
- Use radians on your calculator for $\sin \omega t$ and $\cos \omega t$.
- T is independent of amplitude in SHM; it is damping that makes the amplitude decrease.